Topics: 16.2 Line Integrals (Second Part); 16.3 The Fundamental Theorem for Line Integrals
Please put away all electronic devices, including cell phones and calculators.

1. Evaluate the vector line integral.

   (a) $\displaystyle\int_C y\,dx + 2x\,dy$, where C is parametrized by $x = 3t,\ y = t^2$ with $0 \le t \le 1$.

   We have that
   $$dx = 3\,dt \quad \text{and} \quad dy = 2t\,dt.$$
   Thus,
   $$\begin{aligned}\int_C y\,dx + 2x\,dy &= \int_0^1 \left((t^2)(3\,dt) + 2(3t)(2t\,dt)\right)\\ &= \int_0^1 \left(3t^2 + 12t^2\right)\,dt\\ &= \int_0^1 15t^2\,dt\\ &= 15 \cdot \frac{t^3}{3}\bigg|_0^1\\ &= 5.\end{aligned}$$

   (b) $\displaystyle\int_C x\,dx + y\,dy + x^2y\,dz$, where C is given by $\mathbf{r}(t) = \langle \cos t, \sin t, t\rangle$ with $0 \le t \le \frac{\pi}{2}$.

   We have that
   $$dx = -\sin t\,dt, \qquad dy = \cos t\,dt, \qquad dz = dt.$$
   Thus,
   $$\begin{aligned}\int_C x\,dx + y\,dy + x^2y\,dz &= \int_0^{\pi/2} \left((\cos t)(-\sin t\,dt) + (\sin t)(\cos t\,dt) + (\cos^2 t\sin t)(dt)\right)\\ &= \int_0^{\pi/2} \left(-\cos t\sin t + \sin t\cos t + \cos^2 t\sin t\right)\,dt\\ &= \int_0^{\pi/2} \cos^2 t\sin t\,dt\\ &= -\frac{\cos^3 t}{3}\bigg|_0^{\pi/2}\\ &= \frac{1}{3}.\end{aligned}$$

2. Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F}(x,y) = xy^2\,\mathbf{i} - x^2\mathbf{j}$ and C is given by $\mathbf{r}(t) = t^3\,\mathbf{i} + t^2\,\mathbf{j}$ with $0 \le t \le 1$.

Since $\mathbf{r}(t) = t^3\,\mathbf{i} + t^2\,\mathbf{j}$, we have that

$$d\mathbf{r} = \mathbf{r}'(t)\,dt = \langle 3t^2, 2t\rangle\,dt.$$

Also,

$$\mathbf{F}(\mathbf{r}(t)) = \mathbf{F}(t^3, t^2) = \left\langle t^3(t^2)^2, -(t^3)^2\right\rangle = \langle t^7, -t^6\rangle.$$

Thus,

$$\begin{aligned}
\int_C \mathbf{F}\cdot d\mathbf{r} &= \int_0^1 \langle t^7, -t^6\rangle \cdot \langle 3t^2, 2t\rangle\,dt \\
&= \int_0^1 (3t^9 - 2t^7)\,dt \\
&= \left[\frac{3}{10}t^{10} - \frac{1}{4}t^8\right]_0^1 \\
&= \frac{3}{10} - \frac{1}{4} \\
&= \frac{1}{20}.
\end{aligned}$$

3. Find the work done by the force field

$$\mathbf{F}(x, y) = y^2\,\mathbf{i} - y\sin x\,\mathbf{j}$$

in moving an object along the curve $x = y^2 + 2$ from $(2, 0)$ to $(6, 2)$.

We parametrize the path by $\mathbf{r}(t) = \langle t^2 + 2, t\rangle$ for $0 \le t \le 2$. Differentiating, we find that

$$\mathbf{r}'(t) = \langle 2t, 1\rangle.$$

Now, to find work, we compute the vector line integral,

$$\begin{aligned}
\text{Work} &= \int_C \mathbf{F}\cdot d\mathbf{r} \\
&= \int_0^2 (t^2\,\mathbf{i} - t\sin(t^2+2)\,\mathbf{j}) \cdot (2t\,\mathbf{i} + 1\,\mathbf{j})\,dt \\
&= \int_0^2 2t^3 - t\sin(t^2+2)\,dt \\
&= \frac{1}{2}t^4 + \frac{1}{2}\cos(t^2+2)\Big|_0^2 \\
&= 8 + \frac{1}{2}\cos 6 - \frac{1}{2}\cos 2.
\end{aligned}$$

Thus, the work done is $8 + \frac{1}{2}\cos 6 - \frac{1}{2}\cos 2$ units.

4. Determine whether the vector field $\mathbf{F}$ is conservative. If it is, find a function f with $\mathbf{F} = \nabla f$.

(a) $\mathbf{F}(x,y) = (xy + y^2)\,\mathbf{i} + (x^2 + 2xy)\,\mathbf{j}$

> Here $P(x,y) = xy + y^2$ and $Q(x,y) = x^2 + 2xy$. Then
>
> $$P_y = x + 2y, \qquad Q_x = 2x + 2y.$$
>
> Since $P_y \neq Q_x$, the vector field is not conservative.

(b) $\mathbf{F}(x,y) = ye^x\,\mathbf{i} + (e^x + e^y)\,\mathbf{j}$

> Here $P(x,y) = ye^x$ and $Q(x,y) = e^x + e^y$. Since
>
> $$P_y = e^x = Q_x$$
>
> and the domain is all of $\mathbb{R}^2$, the given vector field is conservative.
> To find a potential function f, we first solve $f_x = ye^x$. Integrating with respect to x, we get
>
> $$f(x,y) = ye^x + g(y),$$
>
> where g is a function of y. Taking the partial derivative of f with respect to y,
>
> $$f_y = e^x + g'(y).$$
>
> Because $f_y = Q = e^x + e^y$, it follows that $g'(y) = e^y$. So $g(y) = e^y + C$, and therefore
>
> $$f(x,y) = ye^x + e^y + C.$$
>
> Thus one function f with $\mathbf{F} = \nabla f$ is $f(x,y) = ye^x + e^y$.

5. (a) Show that the vector field $\mathbf{F}(x,y) = (1 + xy)e^{xy}\,\mathbf{i} + x^2e^{xy}\,\mathbf{j}$ is conservative and find a potential function f for $\mathbf{F}$.

> Here $P(x,y) = (1 + xy)e^{xy}$ and $Q(x,y) = x^2e^{xy}$. We have that
>
> $$P_y = e^{xy}(2x + x^2y), \qquad Q_x = e^{xy}(2x + x^2y).$$
>
> Since $P_y = Q_x$ and the domain is all of $\mathbb{R}^2$, $\mathbf{F}$ is conservative.
> To find a potential function f, we first solve $f_y = x^2e^{xy}$. Integrating with respect to y, we get
>
> $$f(x,y) = xe^{xy} + h(x),$$
>
> where h is a function of x. Differentiating with respect to x, we obtain
>
> $$f_x = e^{xy} + x(ye^{xy}) + h'(x) = (1 + xy)e^{xy} + h'(x).$$
>
> Since $f_x = P = (1 + xy)e^{xy}$, it follows that $h'(x) = 0$. Thus $h(x) = C$, so
>
> $$f(x,y) = xe^{xy} + C.$$
>
> Therefore, one potential function is $f(x,y) = xe^{xy}$.

(b) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is given by $\mathbf{r}(t) = \cos t\,\mathbf{i} + 2\sin t\,\mathbf{j}$ with $0 \leq t \leq \frac{\pi}{2}$.

By the Fundamental Theorem for Line Integrals,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{r}(\pi/2)) - f(\mathbf{r}(0)).$$

We have that

$$\mathbf{r}(0) = (1, 0), \qquad \mathbf{r}(\pi/2) = (0, 2).$$

Evaluating $f(x, y) = xe^{xy}$ at these points,

$$f(0, 2) = 0, \qquad f(1, 0) = 1.$$

Hence

$$\int_C \mathbf{F} \cdot d\mathbf{r} = 0 - 1 = -1.$$

6. Show that the vector line integral $\int_C \sin y\, dx + (x \cos y - \sin y)\, dy$ is independent of path, where C is any piecewise smooth curve from $(2, 0)$ to $(1, \pi)$. Then evaluate the integral.

Here $P(x, y) = \sin y$ and $Q(x, y) = x \cos y - \sin y$. We have that

$$P_y = \cos y, \qquad Q_x = \cos y.$$

Since $P_y = Q_x$ and the domain is all of $\mathbb{R}^2$, the vector field $\mathbf{F} = \langle P, Q \rangle$ is conservative, so the line integral is independent of path.

To find a potential function f, we first solve $f_x = \sin y$. Integrating with respect to x, we get

$$f(x, y) = x \sin y + g(y),$$

where g is a function of y. Differentiating with respect to y, we obtain

$$f_y = x \cos y + g'(y).$$

Since $f_y = Q = x \cos y - \sin y$, it follows that $g'(y) = -\sin y$. Thus $g(y) = \cos y + C$, so

$$f(x, y) = x \sin y + \cos y + C.$$

Therefore, one potential function is $f(x, y) = x \sin y + \cos y$.

Now C is any path from $(2, 0)$ to $(1, \pi)$. By the Fundamental Theorem for Line Integrals,

$$\int_C \sin y\, dx + (x \cos y - \sin y)\, dy = f(1, \pi) - f(2, 0) = -1 - 1 = -2.$$

7. Use the Fundamental Theorem of Line Integrals to find the work done by the force field

$$\mathbf{F}(x, y, z) = (ze^y + \cos y)\, \mathbf{i} + (xze^y - x \sin y)\, \mathbf{j} + (xe^y - 2z)\, \mathbf{k}$$

in moving an object along a piecewise smooth path from $(0, 3, 2)$ to $(1, 0, 2)$.

We have that $\mathbf{F}(x, y, z) = \langle P, Q, R \rangle$, where

$$P = ze^y + \cos y, \qquad Q = xze^y - x\sin y, \qquad R = xe^y - 2z.$$

To use the Fundamental Theorem for Line Integrals, we first find a potential function f such that $\nabla f = \mathbf{F}$. Since $f_x = P$, we integrate with respect to x:

$$f(x, y, z) = xze^y + x\cos y + g(y, z),$$

where g is a function of y and z.

Differentiating with respect to y, we get

$$f_y = xze^y - x\sin y + g_y(y, z).$$

Since $f_y = Q = xze^y - x\sin y$, it follows that $g_y(y, z) = 0$. Thus g depends only on z, so $g(y, z) = h(z)$. Therefore

$$f(x, y, z) = xze^y + x\cos y + h(z).$$

Now differentiate with respect to z:

$$f_z = xe^y + h'(z).$$

Since $f_z = R = xe^y - 2z$, we have $h'(z) = -2z$. Hence $h(z) = -z^2 + C$, so

$$f(x, y, z) = xze^y + x\cos y - z^2 + C.$$

Therefore, one potential function is

$$f(x, y, z) = xze^y + x\cos y - z^2.$$

If C is any piecewise smooth path from $(0, 3, 2)$ to $(1, 0, 2)$, then the work done is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 0, 2) - f(0, 3, 2) = -1 - (-4) = 3.$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Vector Fields. A vector field $\mathbf{F}$ is a **gradient field** if there exists a function f such that

$$\mathbf{F} = \nabla f.$$

In this case, $\mathbf{F}$ is also said to be **conservative**, and f is called a **potential function** for $\mathbf{F}$.

Vector Line Integrals. Let C be a smooth curve given by the vector function $\mathbf{r}$ with $a \leq t \leq b$, and let $\mathbf{F}$ be a continuous vector field. The **vector line integral** of $\mathbf{F}$ along C is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T}\, ds = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt.$$

If $\mathbf{F}(x, y, z) = P(x, y, z)\,\mathbf{i} + Q(x, y, z)\,\mathbf{j} + R(x, y, z)\,\mathbf{k}$, then we also write

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P\, dx + Q\, dy + R\, dz$$

and similarly for a curve and vector field in two dimensions.

Work. The **work** done by a force field $\mathbf{F}$ in moving an object along a curve C is

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

The Fundamental Theorem for Line Integrals. Let C be a piecewise smooth curve given by $\mathbf{r}(t)$, $a \leq t \leq b$, and suppose that $\mathbf{F} = \nabla f$ for some function f. Assume that $\mathbf{F}$ is continuous. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a)).$$

Suggested Textbook Problems
Section 16.2 (Second Part): 5-8, 14-24, 29-32, 34, 41-46
Section 16.3: 1-36, 41